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LAPTOP PRICES PREDICTOR

Introduction

With the rapid growth of the technology market, laptops are available in a wide range of configurations and price segments. Factors such as brand reputation, RAM capacity, storage size, processor generation, and screen size significantly influence the price of a laptop. Determining an appropriate price based on these specifications can be challenging for both consumers and retailers.

Machine learning provides an effective solution for analyzing historical data and identifying patterns that influence pricing. By training predictive models on laptop specifications and corresponding prices, it is possible to estimate the market value of a laptop with reasonable accuracy.

The Laptop Price Predictor project utilizes a Linear Regression algorithm to predict laptop prices based on key hardware specifications. The project was developed using Python and popular data science libraries including Pandas, Scikit-learn, and Matplotlib in the Google Colab environment.

This project demonstrates the practical application of machine learning in the field of e-commerce and consumer electronics, highlighting the role of data-driven decision-making in pricing strategies.

Abstract

The Laptop Price Predictor is a machine learning project developed to estimate laptop prices based on various hardware specifications. The objective of this project is to demonstrate the application of predictive analytics in the consumer electronics domain by using machine learning techniques to forecast laptop prices accurately.

The dataset used in this project contains information about laptop brands, RAM capacity, storage size, processor generation, screen size, and corresponding prices. The dataset was created and processed using Python libraries such as Pandas and NumPy. Since the dataset includes categorical variables, one-hot encoding was applied to convert brand names into numerical values suitable for model training.

A Linear Regression algorithm from the Scikit-learn library was used to train the prediction model. The dataset was divided into training and testing subsets to evaluate the model's performance. Mean Absolute Error (MAE) and R^2 Score metrics were used to measure prediction accuracy.

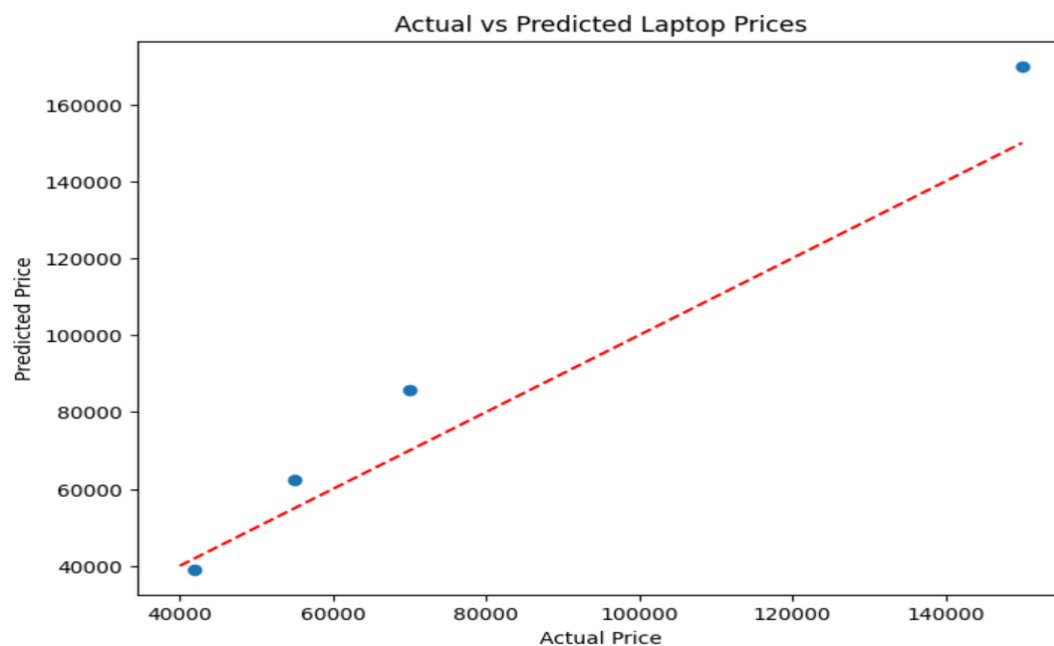
Matplotlib was utilized to visualize the relationship between actual and predicted prices. The results demonstrate that machine learning can effectively estimate laptop prices based on hardware specifications, providing valuable insights for consumers and businesses.

Result and screenshots

	Brand	RAM_GB	Storage_GB	Processor_Gen	Screen_Size	Price
0	Dell	8	256	10	15.6	55000
1	HP	16	512	11	14.0	70000
2	Lenovo	8	512	10	15.6	65000
3	Apple	16	512	12	13.3	120000
4	Asus	8	256	10	15.6	60000

Mean Absolute Error: 11515.16
R² Score: 0.9

Predicted Laptop Price: NPR 83736.47



1. Dataset preview
2. MAE and R² Score
3. Predicted laptop price
4. Graph

Conclusion

The Laptop Price Predictor project successfully demonstrates the application of machine learning techniques for estimating laptop prices based on hardware specifications such as brand, RAM capacity, storage size, processor generation, and screen size. A Linear Regression model was developed and trained using a structured dataset to identify the relationship between these features and laptop prices.

The model was evaluated using performance metrics such as Mean Absolute Error (MAE) and R^2 Score, which indicated satisfactory prediction accuracy. The visualization of actual versus predicted prices further validated the effectiveness of the model.

This project highlights the potential of machine learning in supporting data-driven decision-making within the consumer electronics market. The developed model can assist customers in comparing laptop prices and help retailers estimate pricing based on product specifications.

Future enhancements may include using larger real-world datasets, incorporating additional features such as graphics card specifications, battery life, and processor type, and experimenting with advanced machine learning algorithms to improve prediction accuracy.